

WOMD Recorded Scenario (Retrospective Snapshot t)

- SDC State: (x_0, y_0, v_0, ψ_0)
- Nearest Clearance Focal Actor
- Non-Focal Dynamic Actors ($< 70\text{m}$)
- Static Roadway Polylines

[No Cross-Time Feature Leakage]

Focal Controls (P_{clean} , 12 Feat)

- SDC speed, tangential accel, yaw rate
- Focal pos/vel, clearance, type flags

ODD-Context Proxies (E_{all} , 17 Feat)

- E_{static} (5): Crosswalk, Road Edge
- E_{comp} (6): Density, VRU Prop, Entropy
- E_{interact} (6): Non-focal, SDC-relative

Reference Criticality Target (Y_{crit})

- Linear-extrapolated swept OBB-TTC $\leq 3.0\text{s}$
- Minimum across eligible SDC--actor pairs
- Analytical surrogate (not observed crash)

Shared Kinematic
Primitives (x, y, v)
[Convergent Validity]

Nested Model Controls

- M_P : Baseline (P_{clean})
- $M_{P+E_{\text{interact}}}$: Secondary
- $M_{P+E_{\text{all}}}$: Primary Full
- Dev refit ($N = 15,641$ scenarios)
- Scenario-equal weights ($w_{st} = 1/n_s$)

$\hat{y} = \sigma(f(x))$ Predicted Score

Evaluation:
 ΔAP , Scenario-Equal
Paired Block CI

Sealed Internal Holdout Cohort ($N = 2,804$ scenarios, 255,164 frames) | Paired Scenario-Block Percentile Bootstrap ($B = 1,000$, seed=42)